

NUMERICAL MESOSCALE MODELLING OF MICROSTRUCTURES IN INCONEL 617

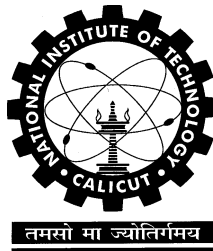
Project Report

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology
in
Mechanical Engineering

by

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CERTIFICATE

This is to certify that the report entitled “**NUMERICAL MESOSCALE MODELLING OF MICROSTRUCTURES IN NICKEL-BASED SUPERALLOY**” is a bonafide record of the **Project** done by **SANDHRA M LAL** (B201054ME), **AISWARYA R** (B201046ME), **THANVIR S DIOUF** (B201120ME), **MAMIDI MAHIMA PRAHARSHA** (B201140ME) under our supervision, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Mechanical** from **National Institute of Technology Calicut**, and this work has not been submitted elsewhere for the award of a degree.

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ABSTRACT

In the past, very little effort was made to explore the formability behavior of high-strength super alloys like Inconel 617 alloy. This project develops a systematic method for the investigation of warm-forming procedures for Inconel 617.

A 2-D standalone CA program was developed in Java to investigate the microstructural evolution during warm forming. The initial grain size, temperature, strain rate, and strain are entered into the 2-D CA model as input data. The simulation's output data include the flow curve, final dynamic recrystallization (DRX) grain diameter (DDRX), and DRX fraction (XDRX). The simulation is run at various strain rates and temperatures by altering the simulation parameters to match the simulation output with the experimental results.

The parameters for work hardening and flow softening were computed. Peak stress and temperature were shown to be related. A relationship between the flow softening parameter and the temperature was also discovered. The mobility parameter was optimized, and the anticipated and experimental findings were confirmed.

Physical Neural Networks (PINNs) analyze how temperature and strain rate affect Inconel 617's behavior during hot working. They incorporate physical equations governing stress-strain rates and grain growth. Trained PINNs predict stress, grain size, and strain, aiding process optimization.

KEYWORDS: Inconel 617, cellular automata, work hardening, flow softening, Physical NeuralNetwork

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LIST OF ABBREVIATIONS

TMP	Thermomechanical processing
WH	Work hardening
DRV	Dynamic recovery
DRX	Dynamic recrystallization
CA	Cellular automata model
DRX	Dynamic Recrystallization Grain Diameter
XDRX	Dynamic Recrystallization Fraction
CAC	Conventional Cellular Automaton
CAM	Modified Cellular Automaton
LR	Linear regression
PINN	Physical Informal Neural Network
TMP	Thermomechanical processing
ANN	Artificial Neural Network

LIST OF SYMBOLS

g	Grain size (mm)
t	Time (s)
ρ	Dislocation density (no. of dislocations/m ²)
μ	Mobility
K_1	Work hardening parameter
K_2	Flow softening parameter
Z	Zenner Hollomon Parameter
$\dot{\epsilon}$	Strain rate (s ⁻¹)
Q_p	Activation Energy (J)
R	Gas Constant = 8.314 J·mol ⁻¹ ·K ⁻¹
T	Temperature (K)
Q_n	Activation energy for nucleation (KJ/mol)
Q_f	Activation energy for diffusion (KJ/mol)
σ_p	Peak Stress (MPa)
G	Shear Modulus (Pa)
α	Constant and the values of K_1 and K_2 can be evaluated from Kocks- Mecking plot

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Superalloys unlike other alloys and metals show distinctive properties. They are high-performance materials known for their exceptional mechanical strength, resistance to high temperatures, corrosion, and oxidation resistance. They also show properties like Thermal creep resistance and Surface stability. These retain all these mechanical properties due to their unique combination of alloying components.

Inconel 617, a nickel-based Superalloy Intrigued us to study it. The remarkable mechanical qualities of this alloy are due to its carefully balanced combination of alloying elements, namely Ni-52.5%, Cr-19%, Fe-17%, and Mo-3.5%. It exhibits remarkable properties like High-temperature resistance and good oxidation and corrosion resistance, which makes it to be used to make catalyst grid support in Chemical industries, and combustion chambers which need high resilience to temperature in gas industries.

Inconel 617 boasts of remarkable properties that make it suitable for a wide range of applications in different industries such as Nuclear, Marine, and Oil. These industries leverage its exceptional features for various purposes.

Desired components for various applications are manufactured by Thermomechanical processing (TMP). Despite its formidable qualities, machining Inconel 617 can pose challenges due to its tendency to work harden quickly, generate excessive heat, and resist metal removal. However, traditional production processes can still be employed with careful consideration of machining parameters.

To create intricate shapes and structures with Inconel 617, thermomechanical processing (TMP) is often utilized. TMP involves manipulating the material's microstructure through controlled temperature and strain interactions. This process is vital for enhancing mechanical properties, but it requires a deep understanding of deformation mechanisms to avoid flow instabilities such as cracks and localization.

The cellular automata (CA) model has emerged as a valuable tool for predicting the behavior of materials like Inconel 617 during thermomechanical processing, particularly in dynamic recrystallization (DRX) scenarios. By carefully adjusting processing parameters such as temperature and strain rates, the workability of Inconel 617 can be optimized, either through dynamic recovery (DRV) or DRX.

In applications ranging from gas turbines to nuclear industries, the performance of Inconel 617 hinges on its microstructure, which can be finely tuned through TMP. However, achieving optimal results requires a thorough understanding of the interplay between microstructure, flow behavior, and processing parameters. Therefore, efforts should focus on correlating these factors to enhance the overall performance and reliability of alloy 617 in various demanding environments.

Physics-Informed Neural Networks (PINNs) are utilized to understand the relationship between process variables (temperature and strain rate) and material behavior during hot working deformation, specifically focusing on Inconel 617. PINNs incorporate fundamental physical equations governing material behavior, including constitutive equations, that relates stress and strain rates, considering temperature, strain rate, and material properties. And grain growth equation which describes how grain size evolves with temperature and strain. During training, PINNs optimize parameters to minimize the difference between predicted and observed values while satisfying physical equations. This enables precise learning of complex, non-linear material behavior. Trained PINNs predict intermediate values of stress, grain size, and strain for given temperature and strain rate inputs. This aids in optimizing hot working processes for improved performance and efficiency.

1.2 PROBLEM DEFINITION

According to a thorough literature review, there is a lot of research on classic metallic alloys including steel, titanium, and aluminum under various testing situations. The past, however, saw very little work put towards understanding the formability behavior of high-strength superalloys like Inconel 617 alloy. This project delves into predicting the warm-forming behavior of Inconel alloy using the conventional 2-D cellular automata model. The predicted result is finally compared with the experimental one.

1.3 OUTLINE OF THE REPORT

The project aims to use a cellular automata model to predict the warm-forming behavior of the Inconel 617 alloy. The work hardening and flow softening parameters were calculated. A relation between peak stress and temperature was developed. An equation relating the flow softening parameter and temperature was also obtained. The mobility parameter was optimized and the predicted results were confirmed with the experimental result.

1.4 MODEL DESCRIPTION

1.4.1 Cellular Automaton

A cellular automaton (CA) is a group of cells structured in a grid of a particular shape, with each cell changing state as a function of time, according to a predetermined set of rules dictated by states of neighboring cells. A CA is discrete in both space and time, which implies that the cells that make up the CA represent one of a finite set of states in each time unit.

Furthermore, the cells in a grid evolve in parallel at discrete time steps by taking into account the statuses of neighboring cells. As a result, the CA evolves in discrete time steps according to a set of rules dependent on the states of neighboring cells. These rules can be implemented iteratively for as many time steps as needed.

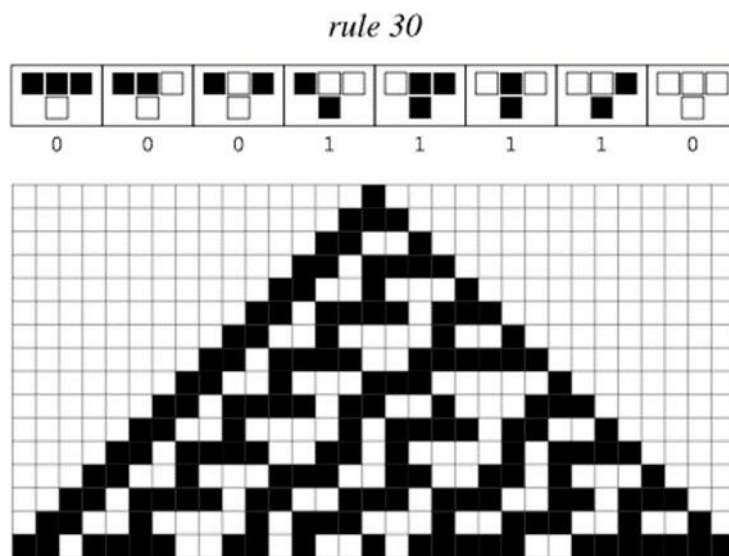


Fig. 1.1: Cellular Automata (<https://mathworld.wolfram.com/CellularAutomaton.html>)

A CA's cells are organized on a grid with a specific shape (square, triangle, hexagon, etc.) and exist in a limited number of dimensions. Every grid cell has a state. While the state has many finite options, the most basic state form is commonly ON or OFF (or TRUE/FALSE or 1/0). The cells next to one another form a cell's neighborhood. Cells in a neighborhood influence one another, and every cell on the CA grid has a neighborhood. Concerning the microstructure of nickel superalloy, new DRX grains nucleate at grain

boundaries, and nucleation and growth events follow the probabilistic CA rules. The growth of each DRX grain is evaluated using a probabilistic approach paired with Moore neighborhoods, which ensures that DRX grains grow circularly.

The flowchart of the CA simulation is as follows:

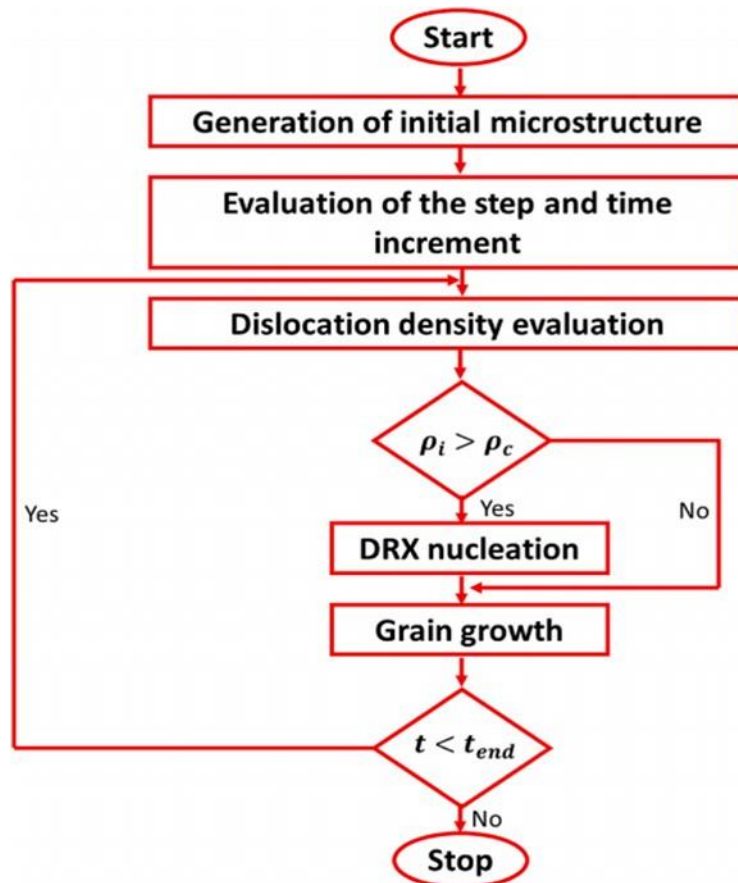


Fig 1.2: Flow chart of CA simulation

1.4.2 Linear regression

It was seen that the regression model approach was deficient in predicting desired parameters. Therefore, the ANN approach was used, and seen that more accurate results were obtained than LR. Regression analysis is a method for determining the connection between dependent variables (response) and independent variables (predictor). In linear regression, the function is a linear equation, and the dependent variable can be written as a function of the independent variable(s) in the form of:

Normalizing equation,

$$X_{norm} = \frac{X - \text{mean}(X)}{\text{std}(X)} \quad (1.1)$$

X represents the input data

Mean(X) is the Mean of input data

Std(X) is the Standard Deviation of input data

Mean Squared error,

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (1.2)$$

N is the number of Data points

y_i , is the Actual output

$\hat{y}_i$, is the predicted output

1.4.3 Artificial Neural Networks

The human brain is an intricately structured, non-linear, and parallel computing system. The motivation for ANN comes from the fundamental structure of the human brain. An artificial neural network (ANN) consists of several interconnected processing components, similar to neurons in the brain.

A model is a black box that contains a set of equations used to calculate the output based on provided input values. A neuron k can be theoretically described using the following equations:

$$u_k = \sum_{j=1}^m W_{kj} x_j \quad (1.3)$$

$$y_k = \varphi(u_k + b_k) \quad (1.4)$$

Bias represented as b_k , alters the net input of the activation function by either increasing or decreasing it.

$x_1, x_2, \dots, x_m$ represent the inputs.

$W_{k1}, W_{k2}, \dots, W_{km}$ represent the weights of neuron k.

u_k represents the output of the linear combiner owing to the input signals.

$\varphi(.)$ represents the activation function.

y_k represents the output signal of the neuron.

The backpropagation method is frequently employed as the learning algorithm for artificial neural networks in multilayered feedforward networks. In backpropagation networks, data undergoes processing as it flows sequentially from the input layer to the hidden layer and finally to the output layer. The objective is to determine the ideal weights that will yield output values close to the desired targets.

Diagram of a neuron:

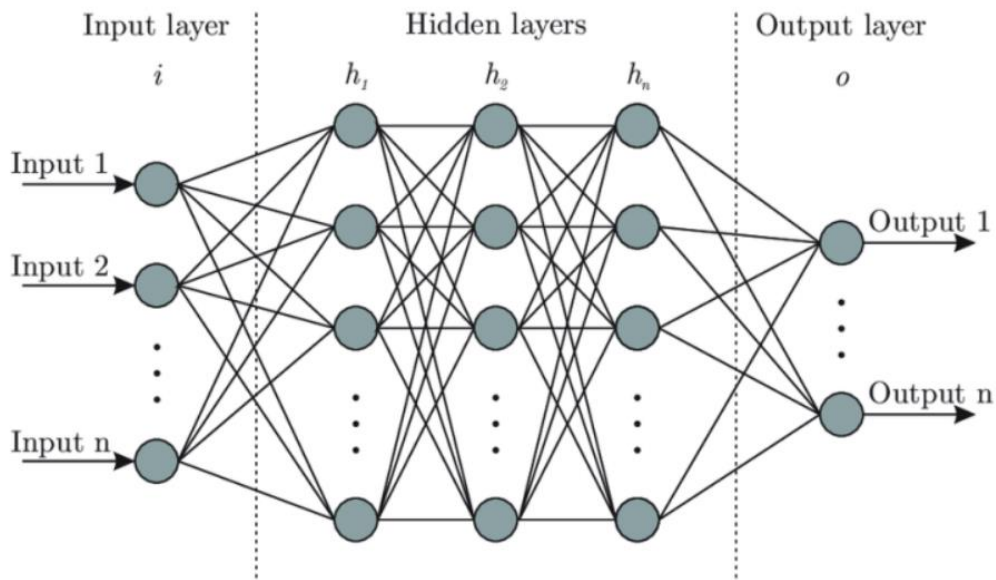


Fig 1.3: Diagram of a Neuron in ANN

1.4.4 Physics Informal Neural Network (PINN)

To produce neural networks that are informed by physics, we take advantage of recent advancements in automated differentiation, one of the most helpful but possibly underutilized approaches in scientific computing, to differentiate neural networks about their input coordinates and model parameters. Such neural networks are conditioned to adhere to any conservation rules, symmetries, or invariances derived from the physical laws governing the observed data, which are represented by general time-dependent, nonlinear partial differential

equations. This straightforward but effective construction allows us to address a broad range of computational science problems. It introduces a potentially revolutionary technology that will enable the creation of new classes of numerical solvers for partial differential equations, data-driven approaches for model inversion, and data-efficient and physics-informed learning machines.

CHAPTER 2

REVIEW OF LITERATURE

2.1 MAJOR CONCLUSIONS OF EACH PAPER

- 1. *Modeling and simulation of dynamic recrystallization in super austenitic stainless steel employing combined cellular automaton, artificial neural network and finite element method., K. Arun Babu, et al (2021):***

CA (Cellular Automaton) models that simulated flow behavior and microstructure evolution of super austenitic stainless steel were developed under various strain rates and temperatures. ANN-based constitutive models accurately estimated flow stress but required numerical procedures for DDRX (Dynamic Recrystallization Grain Diameter) and XDRX (Dynamic Recrystallization Fraction) during FEM simulations. The CAC (Conventional Cellular Automaton) model's representation of microstructure evolution was inadequate due to its failure to consider solute drag effects. CAM (Modified Cellular Automaton) model, addressing this limitation, displayed improved predictive capabilities, accurately predicting microstructural changes and flow behavior with higher correlation coefficients during non-isothermal simulations. CAM model highlighted significant adiabatic heating effects on XDRX at specific temperature-strain rate conditions.

- 2. *2D cellular automaton simulation of hot deformation behavior in a Ni-based superalloy under varying thermal-mechanical conditions., Yan-Xing Liu, et al (2017):***

A 2D cellular automaton (CA) model has been developed effectively to characterize the hot deformation properties of a Ni-based superalloy. This was achieved by establishing relationships between model parameters (such as work hardening parameter, dynamic recovery parameter, nucleation rate, and grain boundary mobility) and thermal-mechanical variables. The CA model demonstrates a high degree of correspondence between its predicted results and experimental findings, thereby confirming its effectiveness in accurately representing the alloy's behavior during hot deformation processes. The CA model approximates outcomes attained under consistent hot deformed conditions when exposed to different thermal-mechanical conditions, which is referred to

as pseudo-metadynamic recrystallization. This phenomenon, known as pseudo-metadynamic recrystallization, refers to a transition in the process of dynamic recrystallization when there is a sudden increase in strain rate. This pause is caused by the inhibition of nucleation and requires additional experimental validation. Unlike traditional finite element simulation methods, the newly developed CA method accurately represents the complex behaviors of hot deformation and the resulting changes in microstructure under various thermal-mechanical conditions.

3. *Characterization of hot deformation behavior of alloy 617 through kinetic analysis, dynamic material modelling, and microstructural studies., K. Arun Babu, et al (2016):*

Mathematical relationships were established between the Zener–Hollomon parameter (Z) and peak stress, as well as between peak strain (ϵ_p) and Z . The average activation energy for the entire hot deformation range of alloy 617 was calculated at 481 KJ mol⁻¹. Dynamic material models were utilized to construct processing maps for alloy 617 at varying strain levels. Strain exhibited minimal impact on the efficiency map, yet higher strain values notably reduced instability at elevated strain rates.⁸⁸ Alloy 617 displayed a broad stable domain at lower strain rates and temperatures, spanning from 1250–1473 K, achieving a peak efficiency of approximately 45%. The identified optimal hot deformation range for alloy 617 was determined as 1323–1423 K and 0.001–0.1 s⁻¹ based on processing maps and microstructural observations.⁸⁸ Within specific strain rate and temperature ranges, an unstable domain revealed heavily deformed grains without a significant fraction of dynamic recrystallization (DRX). A distinctive rise in DRX fraction marked the transition from unstable to stable processing domains.

4. *Study of Grain Growth in a Ni-Based Superalloy by Experiments and Cellular Automaton Model., Yan-Xing Liu, et al (2021):*

The study explores grain growth behavior in a Ni-based superalloy through isothermal heat treatment experiments. Results indicate continuous grain structure coarsening with increased holding time and temperature. A classical grain growth model reveals a predominantly grain boundary migration mechanism, with minor pinning effects from

precipitates. A cellular automaton model is employed to simulate grain growth kinetics, correlating grain boundary velocity with curvature. Grain boundary mobility is found to increase with temperature. Overall, the study provides insights into grain growth mechanisms and offers a model capable of predicting grain size changes under different heat treatment conditions.

5. *Microstructural modelling of dynamic recrystallisation using an extended cellular automaton approach., R. Ding, Z.X. Guo (2002):*

The combination of a Cellular Automaton (CA) model with metallurgical principles provides an excellent simulation of microstructural changes that occur during Dynamic Recrystallization (DRX) in single-phase metallic materials. This model analyzes the impact of processing settings on dynamic recrystallization (DRX) levels and is consistent with experimental findings, especially in a Ti–6Al–4V alloy. It is evident that when the rate of DRX nucleation is minimally affected by the strain rate, consistent microstructures are observed at different combinations of strain rates and temperatures for a specific Zener-Hollomon parameter. However, when the strain rate significantly impacts nucleation, the resulting microstructures vary.

CHAPTER 3

METHODOLOGY

3.1.1 CELLULAR AUTOMATON MODELLING

In material science, work hardening (strain hardening) is the strengthening of metal or polymer by plastic deformation. Work hardening may be desirable, undesirable, or inconsequential depending on the context. Flow stress, during work hardening, increases with the generation and multiplications of dislocations. During dynamic recovery, dislocation gets annihilated, and this variation of dislocation density (ρ) could be expressed by the Kocks-Mecking (KM) relation as follows,

$$\frac{d\rho}{ds} - K_1 \cdot \sqrt{\rho} = \frac{1}{bd} - K_2 \rho \quad (2.1)$$

Throughout the simulation, the dislocation density gets updated based on the KM relation, and the resultant flow stress is evaluated based on the following

$$\sigma = \alpha \mu b \sqrt{\rho} \quad (2.2)$$

The nucleation of new grains occurs when the dislocation density exceeds the value of critical dislocation density. The change in the value of dislocation density after nucleation is not linear, it first drops to zero and then rises to a steady state dislocation density during hardening. The dislocation density from the steady state deformation is used to model the kinetics of dynamic recrystallization during deformation. The dislocation of the deformed matrix then reaches the saturation densities. All of these dislocation densities are evaluated using the flow curves.

In the simulation, the nucleated grains grow to achieve a spherical shape. The energy change associated with this is evaluated based on the following mathematical expressions

$$dW_i = dW_i^S + dW_i^V \quad (2.3)$$

$$dW_i^S = -d(4\pi r_i^2 \gamma_i) = -8\pi r_i \gamma_i dr_i \quad (2.4)$$

$$dW_i^V = 4\pi r_i^2 r(\rho_m - \rho_i) dr_i \quad (2.5)$$

The driving force for boundary migration of the boundary pressure of the i^{th} grain was evaluated using the following relation,

$$F_i = \frac{dW_i}{dr_i} = 4\pi r_i^2 r(\rho_m - \rho_i) - 8\pi r_i \quad (2.6)$$

The value of γ_i was evaluated using the Read-Shockley relationship,

$$\gamma_i = \gamma_m \frac{\theta_i}{\theta_m} \left(1 - \ln \left(\frac{\theta_i}{\theta_m} \right) \right) \quad (2.7)$$

In the model, we are assuming that the grain boundary moves with a velocity, this velocity is given by

$$V = \mu P \quad (2.8)$$

The variation of mobility (μ) with temperature is evaluated using the following

$$\mu = \frac{D_b \times b}{K_2 T} \quad (2.9)$$

Where D_b is Boundary self-diffusion coefficient:

$$D_b = 1.7 \times 10^{-22} \times e^{\left(-\frac{Q_f \times 1000}{RT} \right)} \times e^{Q_f \times 0.091} \quad (2.10)$$

Q_f = Activation energy for diffusion

3.2 MODELING OF WORK HARDENING AND SOFTENING PARAMETERS

The flow stress behavior during Warm-forming could be modeled using the KM relation. The accurate values of k_1 and k_2 are necessary to model the work-hardening behaviors. This was evaluated using the equation

$$K_1 = \frac{K_2 \sigma_p \times 10^6}{\alpha G b} \quad (2.11)$$

Here, b = burger vector = 2.5×10^{-10}

Burgers' vector characterizes the magnitude and direction of lattice distortion associated with a dislocation.

σ_p = Peak Stress

G = Shear Modulus

α is a constant and the values of K_1 and K_2 can be evaluated from Kocks- mecking plot.

The peak stress can be evaluated from the equation,

$$\sigma_p = \frac{1}{0.005439} \ln \left(\sqrt{\left(e^{\frac{\ln(Z)-40.26}{3.86}} \right)^2 + 1} + e^{\frac{\ln(Z)-40.26}{3.86}} \right) \quad (2.12)$$

$$G = 78.9 \times 10^9 \times \left(1 - 0.64 \times \frac{T - 300}{T_m + 273}\right) \quad (2.13)$$

The Zenner Hollomon Parameter Can be evaluated from the equation,

$$Z = \dot{\epsilon} e^{\frac{Q_p}{RT}} \quad (2.14)$$

Z = Zenner Hollomon Parameter

$\dot{\epsilon}$ = Strain rate

Q_p = Activation Energy

R = Gas Constant = $8.314 \text{ J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$

T = Temperature

The variation of $\ln k_2$ was plotted against $\ln Z$ which yields the following-

$$\ln (K_2) = 8.9 - 0.15(\ln (Z)) \quad (2.15)$$

Nucleation Rate,

The nucleation rate is a convenient synthesis of terms that describes how many nuclei of critical size form on a substrate per unit area, per unit time.

It is obtained from the equation,

$$\dot{N} = 3.83 \times 10^{15} \times e^{\left(-\frac{Q_n \times 1000}{RT}\right)} \times \left(\frac{\dot{\epsilon}}{0.2}\right)^{n_p} \quad (2.16)$$

Q_n = activation energy for nucleation.

Dislocation Energy,

$$E_b = \alpha G b^2 \quad (2.17)$$

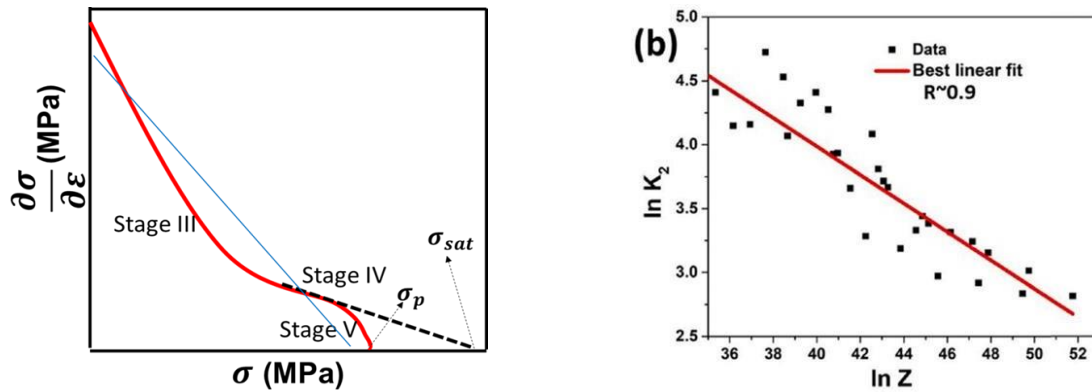


Fig 3.1. (a) Schematic illustration of Kocks-Mecking plot, (b) variation of $\ln K_2$ vs $\ln Z$

CHAPTER 4

WORK DONE

4.1 Briefing on Work of Last Semester

A 2-D standalone CA program was created and modified using Java to investigate the evolution of microstructural evolution during warm forming. The initial grain size, temperature, strain rate, and strain are set as the input data.

4.1.1 *For constant temperature and strain rate*

The values of k_1 and k_2 at a fixed temperature and strain rate are calculated employing the Zener- Holmon parameter. The Java simulation uses these values to represent the behaviors of the superalloy during deformation. Algorithms and rules within the cellular automata are developed to mimic the material's response to plastic deformation. The CA model divides the material into discrete cells, each representing a portion of the material microstructure, and updates their states based on a set of predefined rules. Upon executing the CA model, the evolution of the grain structure is observed over time, capturing the dynamic changes in the material's microstructure. The significant patterns observed in the final grain structures are summarized

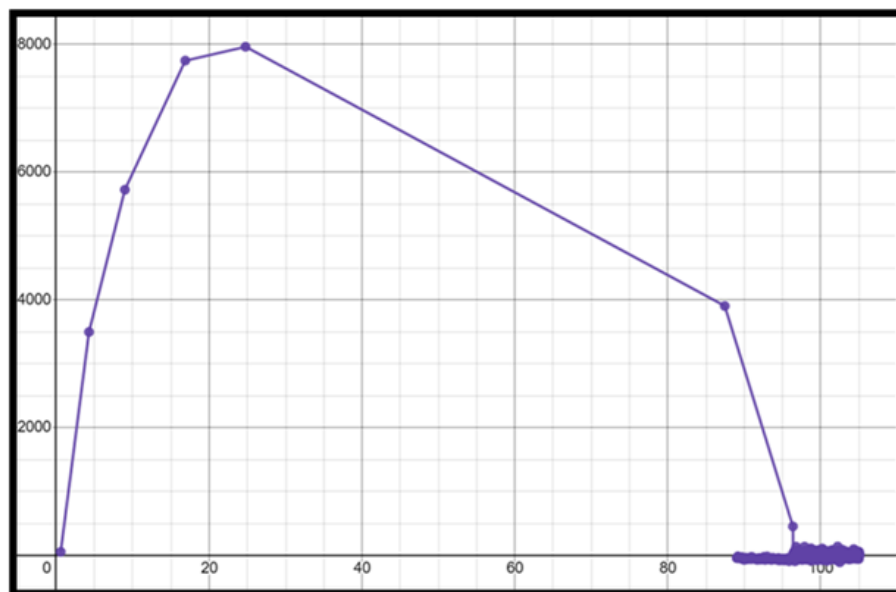
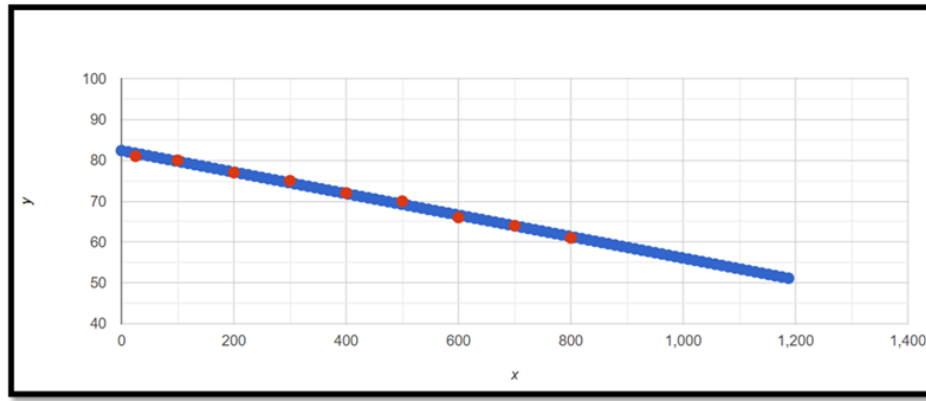


Fig 4.1. Obtained Kocks-Mecking Plot Fig 4.2: Obtained variation of $\ln K_2$ vs. $\ln Z$



4.1.2.1 For constant strain and varying temperature

Work hardening and flow softening parameters are then obtained at a fixed strain rate and a specified temperature range. To establish a precise mathematical relationship between the flow softening parameter (k_2) and temperature (T) variations, curve fitting methods are applied.

Curve fitting involves finding the best-fitting curve that represents the experimental data points. This mathematical approach helps create a functional form that describes how the flow softening parameter changes with temperature. Common techniques include least squares fitting, which minimizes the difference between the predicted and observed values, enabling the derivation of an equation expressing the relationship between k_2 and temperature.

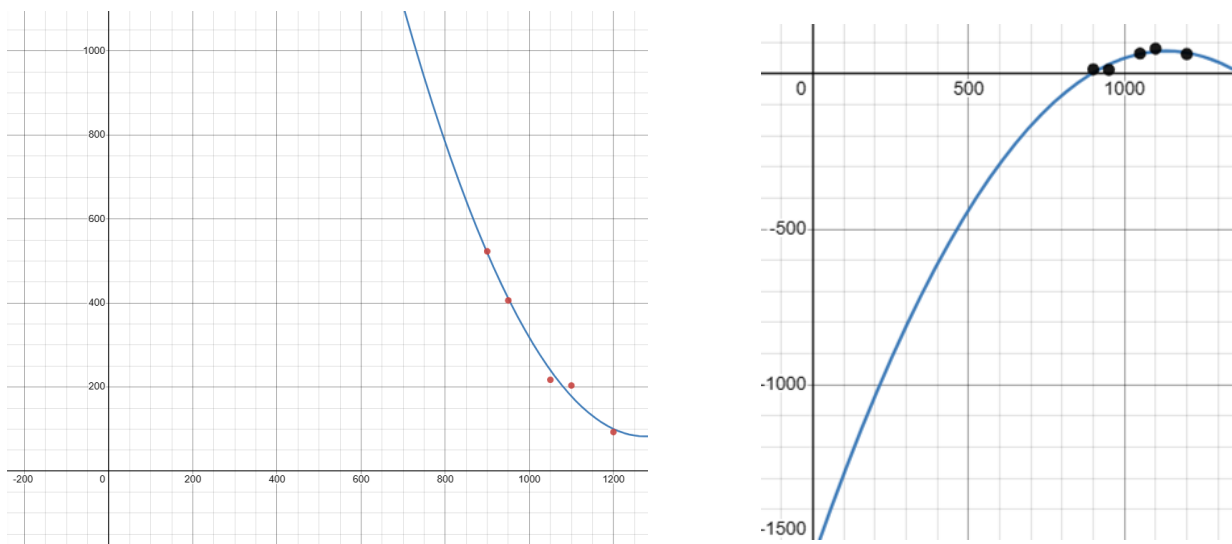


Fig 4.3: Curve Fitting of σ_p vs T and Curve Fitting of k_2 vs T

Similarly, curve fitting is done between peak stress (σ_p) and temperature (T) to derive an equation that accurately describes how peak stress varies with temperature. This provides valuable insight into the material's strength at different temperature conditions.

In the simulation program, an important aspect is aligning the output with existing empirical data. To achieve this alignment, the mobility parameter is adjusted. The mobility parameter affects the movement of dislocations within the material, influencing its plastic deformation behavior.

All these steps collectively contribute to a comprehensive understanding of materials behavior under varying temperature conditions.

The final output from this code is:



Fig 4.4: Pictorial output of microstructures

Graphical representation of the Excel output:

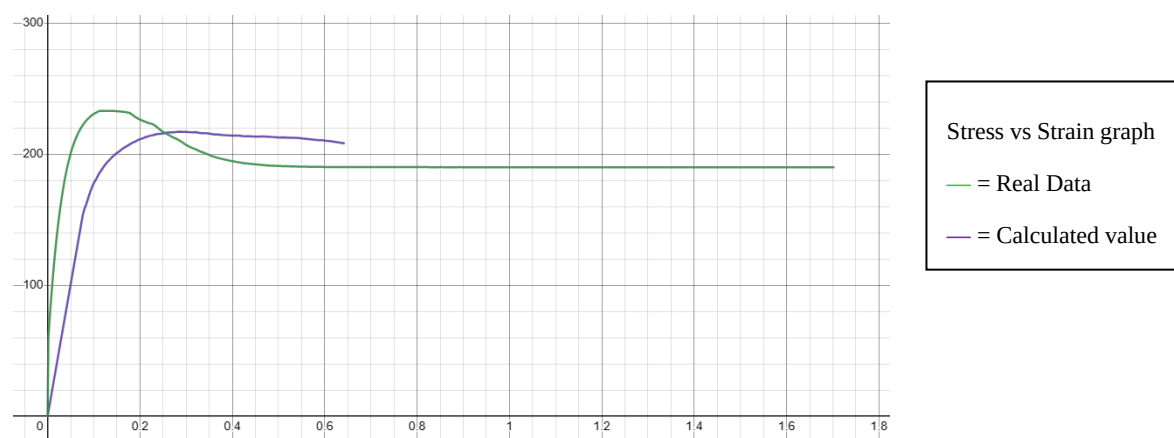


Fig 4.5: Stress vs Strain Graph: Experimental and Calculated

4.2 BRIEFING ON FINAL WORK DONE

4.2.1 Incorporating temperature dependency in mobility and nucleation rate

Instead of using constant values, mobility, and nucleation rate is now determined by an equation that encapsulates its reliance on temperature. This dependency is expressed as a function of the following parameters, boundary thickness (δ), boundary self-diffusion coefficient (D_{ob}), and activation energy (Q_b).

4.2.2 Experimentation for enhancing the accuracy

Few trials were conducted to obtain the optimal value for the product δD_{ob} . This enabled us to fine-tune the simulation model, ensuring it reflects real-world behavior more accurately.

4.2.3 Temperature effects on peak stress and flow softening parameters

The Peak stress (σ_p) and flow softening parameter (K_2) variations with temperature were graphed.

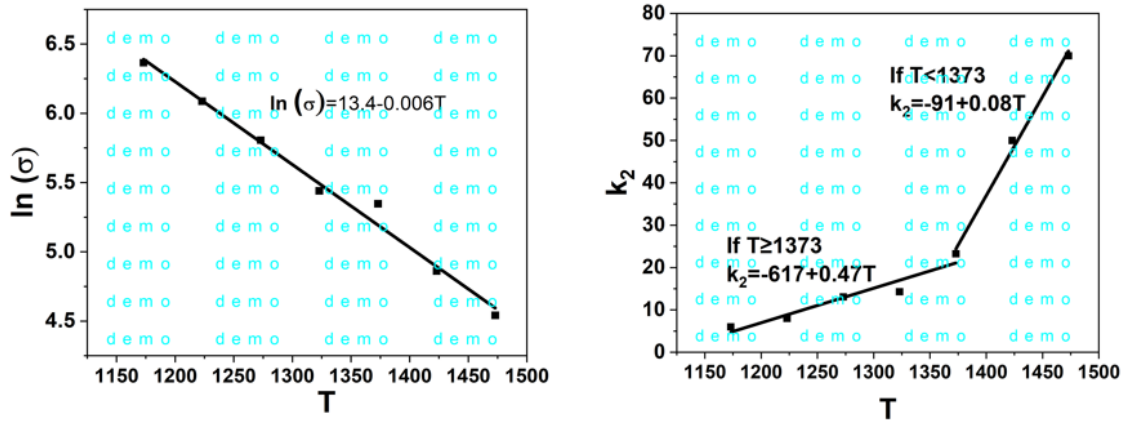


Fig 4.6: A plot between peak stress vs temperature and flow softening parameter vs Temperature

4.2.4 Relationship between the grain diameter and the side length of the cell

The relation between the grain diameter (DDRX, in μm), and length of the cell (LCA, in μm) was observed at a temperature of 1323 K and strain rate 0.2s^{-1}

The polynomial fitting was done using Origin software.

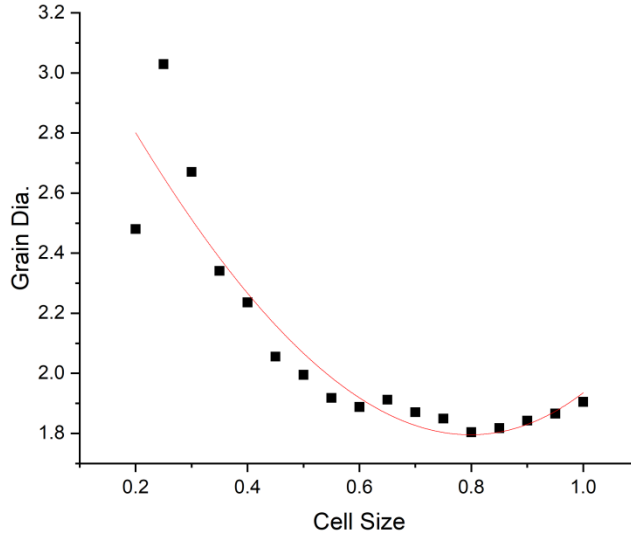


Fig 4.7: A plot between cell size vs Grain diameter

4.2.5 Development of a linear regression model

A model was developed to enhance the accuracy of the CA model. It incorporates the temperature, average grain diameter (DDR_X), and recrystallization fraction (XDR_X). Training of the model involved utilizing the data obtained from both the CA model and experimental results for dependent variables, and Predicted results at intermediate temperatures were presented in a DataFrame and graphically visualized. The mean square error and the coefficient of regression were computed to assess the model's performance.

4.3 EXTRACTION OF IPF DATA

The Inverse Pole Figure (IPF) map of a specimen of Inconel 617 undergoing deformation at different temperatures and strain rate are available from experimental data. The images generated from the simulation are in the dimension 400 x 400 ($\mu\text{m} \times \mu\text{m}$). So, the experimental IPF maps were also cropped to that size to be able to make reasonable comparisons.

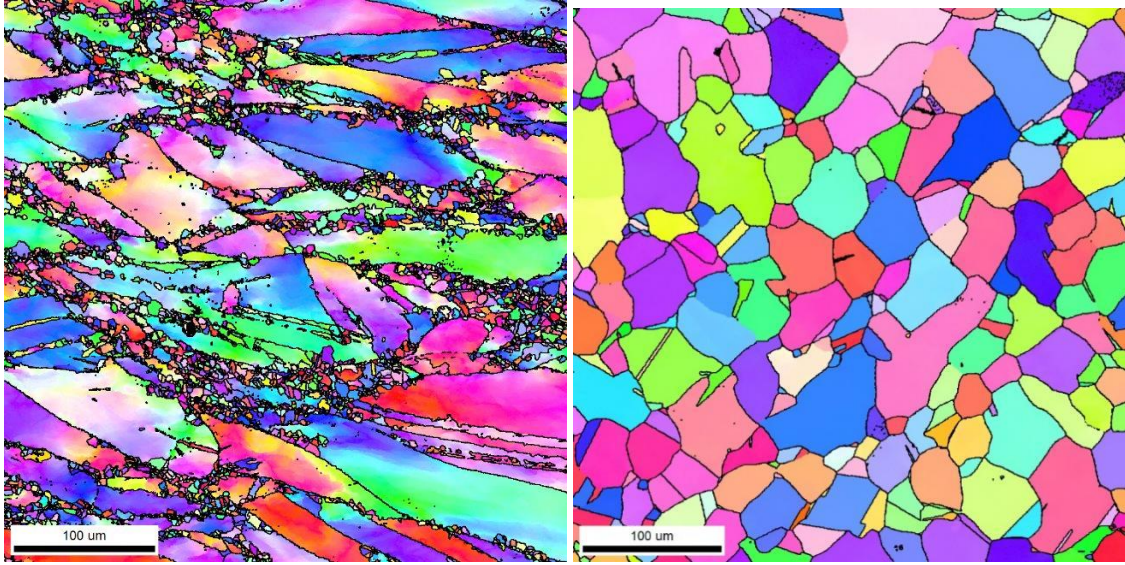


Fig.4.8: IPF map of deformed specimen at 950°C 0.001 s⁻¹
b) 1200°C 0.1 s⁻¹ Size is 400x400(μm x μm)

4.4 CALCULATION OF MORE PARAMETERS

From the reference paper [5] we have obtained the equation:

$$n \ln(\sinh(\alpha\sigma)) = \ln \dot{\epsilon} + \frac{Q}{RT} - \ln A$$

So, to find the parameters, we have plotted these graphs:

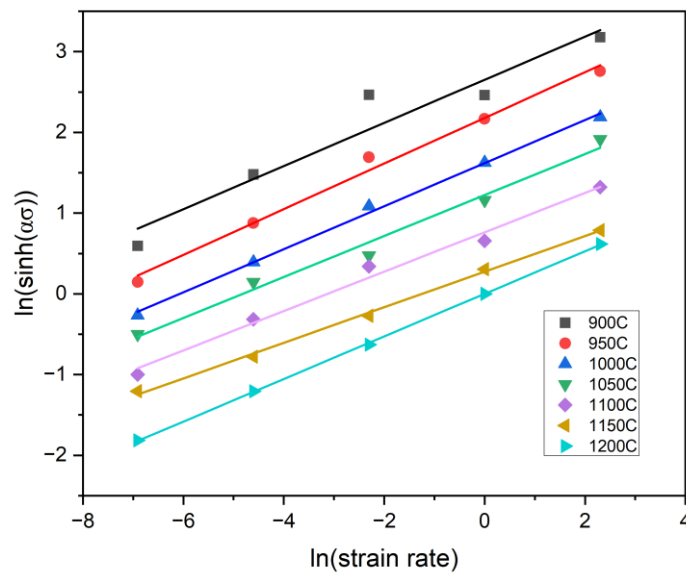


Fig. 4.9: A graph of $\ln(\sinh(\alpha\sigma))$ vs $\ln(\dot{\epsilon})$

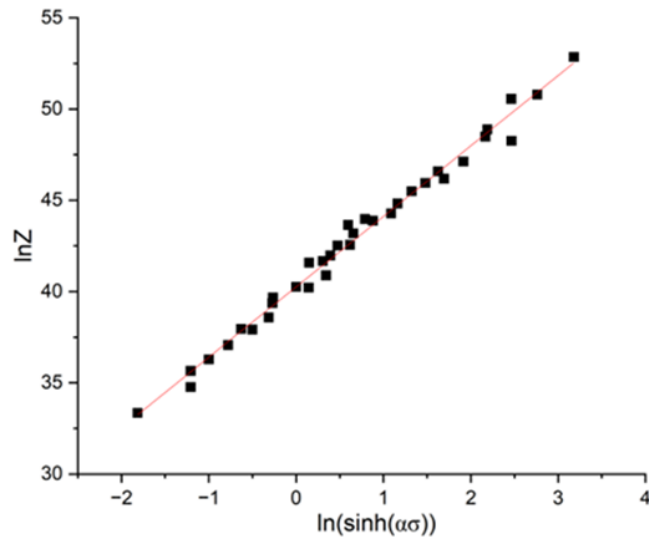


Fig. 4.10: Plot of $\ln Z$ vs $\ln[\sinh(\sigma\alpha)]$

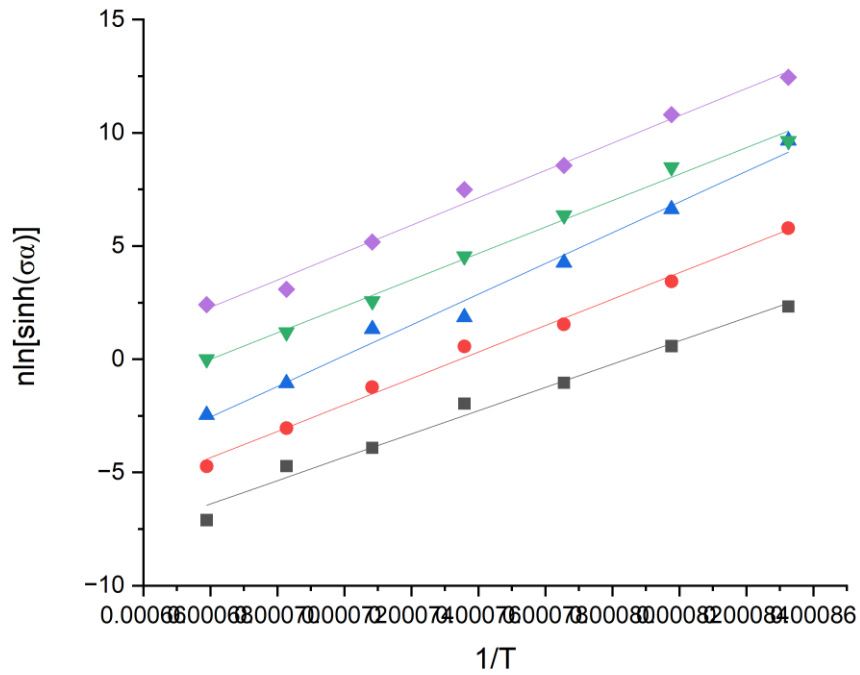


Fig4.11: Plot of $n \cdot \ln[\sinh(\sigma\alpha)]$ vs $1/T$

$$Q=493,039 \text{ J mol}^{-1}$$

$$n=3.91342$$

$$\alpha=0.005439 \text{ Pa}^{-1}$$

4.5 MAKING OF THE NEURAL NETWORK

The making of the neural network to calculate intermediate values was a very straightforward process. Python has a lot of libraries that assist massively in big data processing. Libraries like NumPy and TensorFlow have very well-optimized functions that evaluate data quickly. The data from the CA simulation was loaded into the Python program. The values were then discretized and organized into their corresponding columns. The input features were then normalized using a standard scaler and mesh grids were generated. Then the PINN model was then defined relating the input and output values. Compiling and training the model and single command processes. The program then evaluates the minimum, maximum and normalized values. This is the general process that takes place inside the program when we try to predict an intermediate value.

4.5.1 *Physics-Informed Neural Networks (PINNs) integrate:*

Physics-Informed Neural Networks (PINNs) integrate physical principles into the training process of neural networks to capture complex relationships between input and output data. In the context of hot working deformation of materials like Inconel 617, PINNs are employed to understand the correlations between process variables (such as temperature and strain rate) and material behavior.

The PINN utilizes fundamental physical equations governing the behavior of the material during hot working deformation including the Constitutive Equation, that relates the rate of stress to the rate of strain. It characterizes how the material responds to external forces during deformation, incorporating factors like temperature, strain rate, and material properties. Grain Growth Equation is a crucial aspect of material behavior during deformation processes. The grain growth equation describes how the size and distribution of grains within the material evolve, influenced by temperature and strain. By incorporating these physical equations into the training process, the PINN can effectively learn the intricate non-linear behavior of the material. This means that the neural network not only learns from the input-output data but also respects the underlying physics governing the material's behavior.

During training, the PINN iteratively adjusts its parameters to minimize the discrepancy between predicted and observed values while satisfying the physical equations.

This enables the network to capture complex relationships between process variables and material behavior.

Once trained, the PINN can predict intermediate values of key parameters, such as stress, grain size, and strain, given specific temperature and strain rate inputs. These predictions provide valuable insights into how the material will behave under different processing conditions, facilitating optimization of hot working processes for enhanced performance and efficiency.

CHAPTER 5

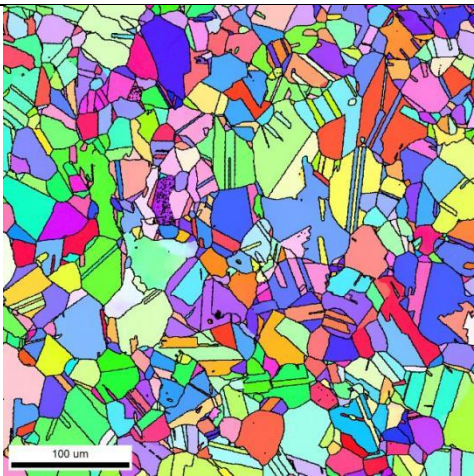
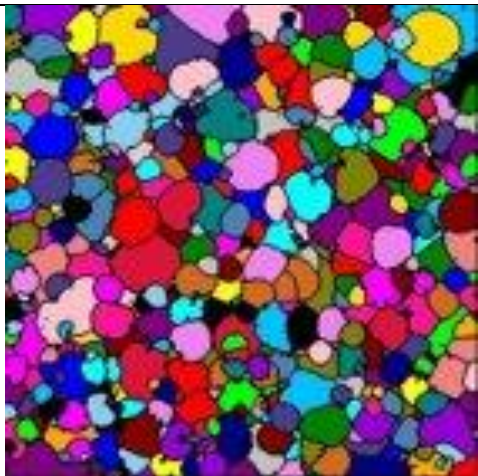
RESULTS

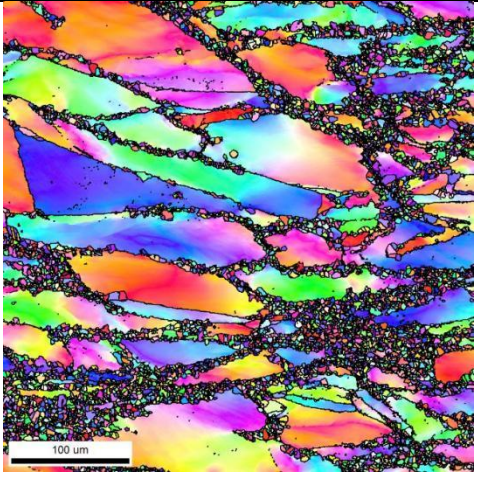
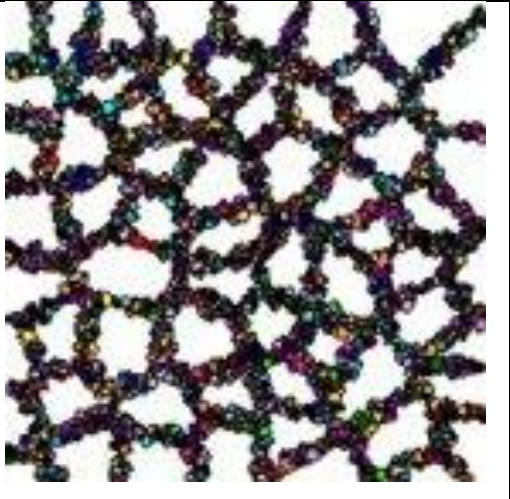
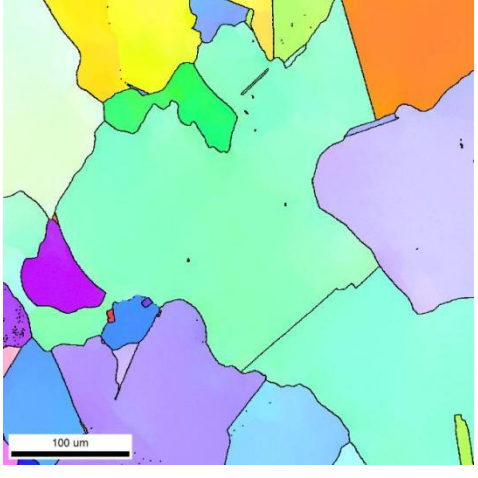
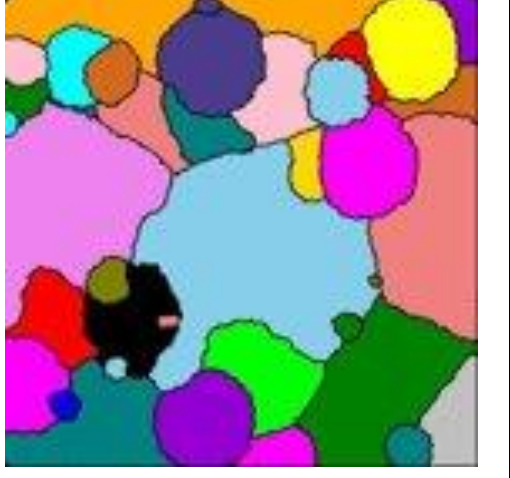
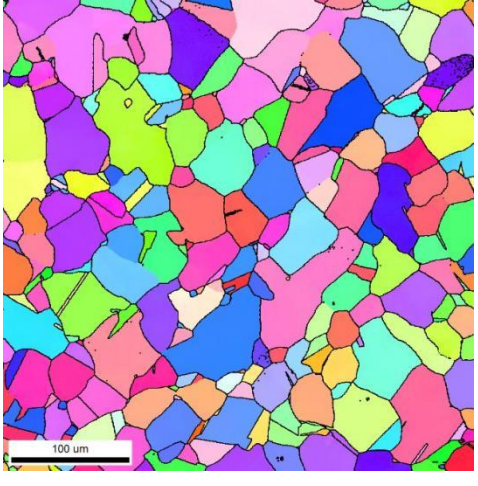
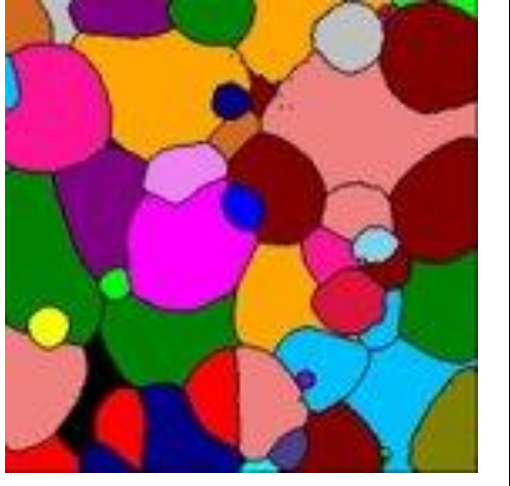
5.1 LARGE DATASET

The cellular automata-based simulation was able to generate a large dataset relating values of flow stress, grain size, and grain fraction across a range of temperatures, strain rate, and strain. The usefulness of this simulation can be attested to the fact that this simulation takes into consideration the processes of work hardening, dynamic recrystallization, and dynamic recovery. The simulation also worked at a microstructural level since it also emulated the grains, their orientation, the movement of dislocations, nucleation, etc.

5.2 IMAGE GENERATION

The simulation was able to generate images that were predicted using the parameters given to it. This makes it easier for us to compare the simulation with experimental data (IPF maps).

Temperature and Strain rate	Experimental	Simulation
1050 °C 0.001 s ⁻¹		

1050 °C 0.1 s⁻¹		
1200 °C 0.001 s⁻¹		
1200 °C 0.1 s⁻¹		

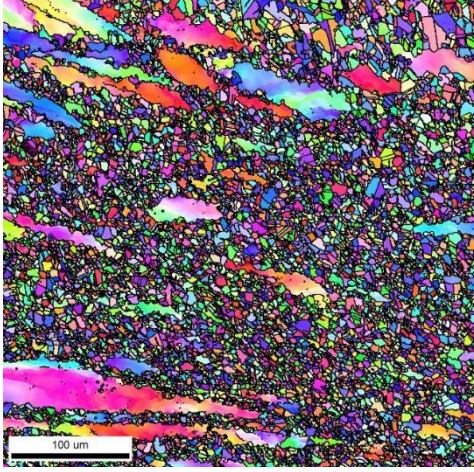
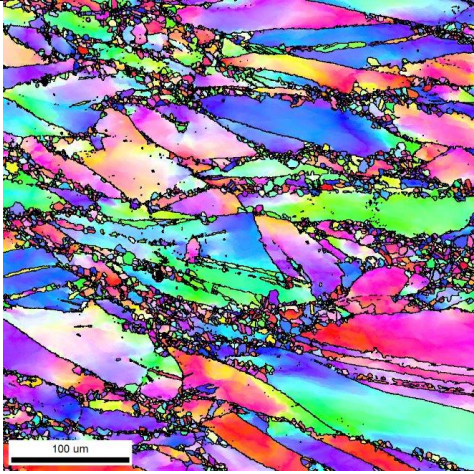
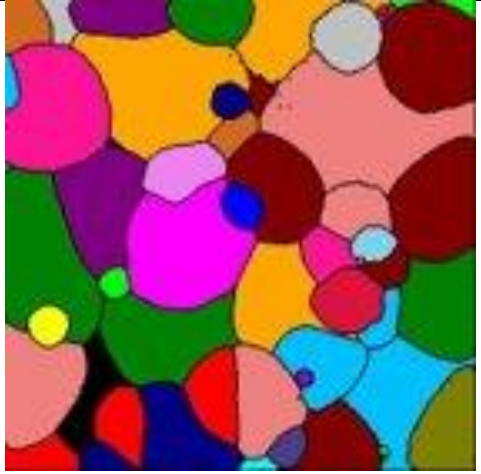
1100 °C 0.1 s ⁻¹		
950 °C 0.001 s ⁻¹		

Table 5.1: Generated images of experimental data

5.3 ARTIFICIAL NEURAL NETWORK

An ANN model was also made using Python and TensorFlow. It is capable of reading the large dataset generated in the simulation and can predict intermediate values using a method called Physics Informed Neural Network (PINN). The section of input dataset is shown along with its corresponding output in Fig.5.1 and Fig 5.2.

150	0.51405	31.805007	1.7847334	8.05E-04	1373	-0.05
151	0.5175	31.035114	1.9456492	9.46E-04	1373	-0.05
152	0.52095	30.309055	1.9546439	0.0011617	1373	-0.05
153	0.5244	29.637186	2.0907508	0.0012984	1373	-0.05
154	0.52785	28.976758	2.1315557	0.0015773	1373	-0.05
155	0.5313	28.391991	2.3064194	0.0017875	1373	-0.05
156	0.53475	27.801793	2.3997535	0.0020586	1373	-0.05
157	0.5382	27.254658	2.499465	0.002325	1373	-0.05
158	0.54165	26.742005	2.5348508	0.0026867	1373	-0.05
159	0.5451	26.440846	2.6368166	0.0029438	1373	-0.05
160	0.54855	25.955929	2.702608	0.0033762	1373	-0.05
161	0.552	25.514016	2.852553	0.0037336	1373	-0.05
162	0.55545	25.080438	2.8983456	0.0041652	1373	-0.05
163	0.5589	24.664063	3.0439087	0.0045746	1373	-0.05
164	0.56235	24.27968	3.1389501	0.0051016	1373	-0.05
165	0.5658	23.90406	3.2261216	0.0055469	1373	-0.05
166	0.56925	23.550509	3.2467979	0.0060852	1373	-0.05
167	0.5727	23.204718	3.3566714	0.0066098	1373	-0.05
168	0.57615	22.876523	3.5057479	0.0072004	1373	-0.05
169	0.5796	22.563855	3.6189464	0.0077723	1373	-0.05
170	0.58305	22.261181	3.6920579	0.0083398	1373	-0.05
171	0.5865	21.955075	3.7698524	0.0090367	1373	-0.05
172	0.58995	21.695671	3.708881	0.0096648	1373	-0.05
173	0.5934	21.423336	3.8368272	0.0103621	1373	-0.05
174	0.59685	21.172186	3.8811192	0.0110258	1373	-0.05
175	0.6003	20.919526	3.9439696	0.0118504	1373	-0.05
176	0.60375	20.591586	3.9551022	0.0125379	1373	-0.05
177	0.6072	20.354575	4.0060578	0.0134223	1373	-0.05
178	0.61065	20.054121	4.1317108	0.0141684	1373	-0.05
179	0.6141	19.839776	4.2748431	0.0150891	1373	-0.05
180	0.61755	19.652467	4.3921309	0.0159516	1373	-0.05
181	0.621	19.464768	4.3696022	0.0169172	1373	-0.05
182	0.62445	19.288313	4.4821111	0.017798	1373	-0.05
183	0.6279	19.107558	4.5132287	0.0189316	1373	-0.05
184	0.63135	18.958761	4.6155163	0.0198133	1373	-0.05
185	0.6348	18.783243	4.7510045	0.0209535	1373	-0.05
186	0.63825	18.554787	4.840026	0.0220457	1373	-0.05
187	0.6417	18.487062	4.9112127	0.0231648	1373	-0.05
188	0.64515	18.342011	4.9755605	0.0242383	1373	-0.05
189	0.6486	18.12085	5.0394341	0.0255359	1373	-0.05
190	0.65205	17.994334	5.0621137	0.0265961	1373	-0.05
191	0.6555	17.782088	5.088313	0.0279086	1373	-0.05
192	0.65895	17.661761	5.1899253	0.0291691	1373	-0.05
193	0.6624	17.617091	5.3143087	0.0304469	1373	-0.05
194	0.66585	17.497823	5.3738298	0.0317137	1373	-0.05
195	0.6693	17.385514	5.4103333	0.0331012	1373	-0.05
196	0.67275	17.341374	5.472746	0.0343953	1373	-0.05
197	0.6762	17.237252	5.4021627	0.0357664	1373	-0.05
198	0.67965	17.128679	5.4326842	0.0371953	1373	-0.05
199	0.6831	17.028199	5.5522881	0.0386152	1373	-0.05
200	0.68655	16.930412	5.6682664	0.039993	1373	-0.05

Fig.5.1. The input of the PINN code for rows 150 to 200

```

Predicted values of drx_gs (rows 150 to 200):
[2.0223823 2.1216207 2.2194312 2.3077772 2.3968034 2.4820457 2.5636208
2.6195889 2.7013345 2.7772434 2.8519652 2.9244032 2.9935892 3.059877
3.1198184 3.1788921 3.2359984 3.291373 3.3532784 3.421196 3.4829915
3.5464845 3.6072013 3.6681154 3.7387338 3.8027725 3.8756099 3.9360573
3.9927979 4.051477 4.1100097 4.16718 4.219204 4.274829 4.3375897
4.3786902 4.4302087 4.4919868 4.5405893 4.6000924 4.647422 4.684722
4.731911 4.7781754 4.8154106 4.8605905 4.90636 4.9571404 5.0131035
5.0676003]
Predicted values of xdrx (rows 150 to 200):
[0.00647534 0.00720183 0.00791822 0.00856807 0.00922275 0.00986459
0.01048712 0.01094134 0.01160889 0.01223077 0.01284336 0.01344165
0.01403324 0.01459935 0.01511027 0.01561432 0.01610292 0.01657785
0.01723635 0.01803944 0.01877873 0.01953575 0.02026382 0.02099396
0.02182701 0.02268111 0.02365016 0.02448205 0.02527049 0.02614431
0.02709411 0.02803715 0.0289126 0.02983914 0.03086689 0.03158733
0.03245559 0.03346943 0.03434022 0.03543371 0.03633263 0.03707117
0.03796778 0.03884962 0.03958705 0.04045158 0.04132551 0.04242394
0.04378986 0.04524033]

```

Fig. 5.2 : Final Output of PINN code

CHAPTER 5

CONCLUSIONS AND SCOPE FOR FUTURE WORK

5.1 CONCLUSION

The following conclusions were made in the present study:

1. A conventional 2-D cellular automata model to predict the microstructural behavior of Inconel 617 alloy under warm working conditions was implemented in Java and its parameters like a hardening parameter, softening parameter, and mobility were modified to best match the experimental results.
2. A relationship between peak stress vs. temperature and flow softening parameter vs. Temperature were established.
3. A PINN was trained to determine the correlations between the process variables and the material's behavior during hot working deformation.
4. For a given set of temperature and strain rate parameters, the trained PINN can forecast the intermediate values of stress, grain size, and strain.

5.2 FUTURE SCOPE

The results generated from this project can be a starting point for future researchers in many ways:

1. They could enter the data from this project into FEM software and simulate the deformation of structures of different shapes and at different temperatures and other conditions. FEM simulations will be able to bridge the gap between grain-scale and component-scale simulations.
2. Future researchers might benefit from the methods we employed in this project. The CA simulation written in JAVA has a lot of solutions to problems that we faced while trying to perfectly simulate the hardening and softening of the superalloy. A lot of logical loops and conditional statements were used to circumvent errors we faced in earlier versions of the code.

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